

Milestone | YouTube Trending Analysis

- All of your answers should be typed or pasted into the grey boxes.
- Questions with grey answer boxes expect a specific answer. Once you input the correct answer, the question number box **will change from blue to green**.
- Questions with yellow answer boxes are free-response questions. There will be no change in the question number boxes (in blue), regardless of your answer.
- Hover your cursor over the blue hint box to get a helpful hint!

Part 1: Understanding the Data

For all the questions below, you'll refer to the dataset in the YouTubeVideos Dataset sheet. The table is named YouTubeVideos .		
1	How many columns take on numeric values?	7
2	Are there any numeric columns that should NOT be treated like standard numbers? If so, which ones are these? An example of this would be phone numbers. Yes, they are numbers, but it doesn't make sense to take the average or their sum, the result would	There are numeric columns that should not be treated like standard quantitative values. These include Trending Date and Publish Time (these represent temporal information rather than measurable magnitude), and Category ID is a numeric identifier used for labeling categories and not for mathematical analysis.
Prompt ChatGPT: Can you explain why category IDs and datetime values (like trending_date and publish_time) shouldn't be treated like regular numbers in data analysis? Provide an example relevant to YouTube data.		
3	How many columns take on text values?	5
4	Are there any text columns that have some natural ordering that lets them be interpreted as more than just text? If so, which ones are these? An example of natural ordering is t-shirt sizes. They are represented by text values but S < M < L < XL	There does not appear to be any text-based columns with a natural ordering. Fields such as video titles, channel names, and category names are categorical labels rather than ordered values.
5	How many columns take on boolean values? Remember: Boolean values are anything represented as True / False	3

Hint!

Part 2: Exploring Key Metrics

Now it's time to look into the actual data. Use functions to answer the following questions. Remember, the table is named YouTubeVideos .		
6	How many videos are there in the dataset?	6,351
7	What is the total number of likes across all trending videos in the dataset?	352,957,165
8	What is the average number of views on a trending video in this dataset?	1,962,117
Use sorting and filtering to help you answer the following questions:		
9	How many trending videos were published before 2010? (Do not include 2010 in your filter.)	9
10	What is the name of the video with the most views that is NOT from the Music or Entertainment categories?	To Our Daughter
11	How many videos in the Gaming category have more than 10 million views?	5
12	How many videos in the Nonprofits & Activism category have more than the average number of views?	2

Part 3: Defining Exceptional Videos

This final question does not require any specific parts of the data, but you might want to use the chart and summary table on the right to help you formulate a response! Remember to <u>unfilter the data</u> first!		
13	Based on your analysis of view counts in trending videos, and what you notice in the summary table to the right, use ChatGPT to help you determine whether a video with 2 million views should be considered exceptional. Ask ChatGPT to help you evaluate the significance of this view count, considering factors like video category, publish date, and other potential influencing variables. What does ChatGPT suggest about defining 'exceptional' in this context, and do you agree with its reasoning?	

Use this summary table for question 13

Category	Number of videos	Average of views
Autos & Vehicles	71	1449487
Comedy	544	1422507
Education	251	607885
Entertainment	1621	1744879
Film & Animation	320	2545387
Gaming	103	2325087
Howto & Style	594	937806
Music	801	6027683
News & Politics	504	459931
Nonprofits & Activism	14	2218901
People & Blogs	495	1348809
Pets & Animals	140	744319
Science & Technology	381	1124489
Shows	4	762400
Sports	449	1424029
Travel & Events	59	996254

What does ChatGPT suggest about defining 'exceptional' in this context, and do you agree with its reasoning?

After reviewing the view count analysis and the summary table, ChatGPT suggests that a video with 2 million views should not automatically be considered exceptional without additional context. The significance of this number depends largely on factors such as video category, timing, and the typical performance within each category. Based on the summary table, categories such as Music and Entertainment consistently average much higher view counts, making 2 million views relatively common in those categories. However, categories like Education tend to have much lower average view counts, meaning a video with 2 million views in that category would stand out more clearly.

This demonstrates that evaluating a video in isolation can be misleading without considering category-level benchmarks. ChatGPT suggests that content should be defined as exceptional based on its relative performance, rather than a fixed view count. I agree with this reasoning, as YouTube videos are best evaluated by comparing them to similar content within the same category and timeframe.

14 Based on your analysis, what key factors do you believe most influence a video's likelihood of trending on YouTube? How would you recommend a client adjust their content strategy to maximize reach and engagement based on your findings? Remember you can use ChatGPT as your teammate in helping to interpret these results in the context of your overall goal.

believe some of the key factors that influence whether a video trends are related to existing audience size and content presentation. Creators with established audiences, such as well-known music artists, are more likely to see strong initial performance because they already have a large and engaged viewer base. For example, an artist like Bruno Mars is likely to generate significantly more views shortly after release than someone who is just beginning their career, which increases the chances of the video trending. Another important factor is visual presentation, particularly thumbnails. Thumbnails play a major role in attracting clicks, and data suggests that more visually appealing thumbnails can

LevelUp: Defining Additional Data

15 Based on what you've analyzed so far, what other types of data or additional variables do you think could help you better understand the factors that drive views on trending videos? Use ChatGPT as your teammate to brainstorm other potential data sources or attributes you could include in your analysis, and explain why they might add value to your findings

Prompt ChatGPT: I'm analyzing trending videos based on likes and views. What other types of data or variables would be helpful for understanding why certain videos perform better? How can I incorporate this new information into my existing analysis to maximize content's reach and engagement.

To better understand why certain videos receive higher view counts, additional data would be helpful. One important factor would be watch time or average view duration, since videos that keep viewers watching longer are more likely to perform better. Engagement metrics such as likes and comments would also be useful, as higher engagement usually indicates stronger audience interest.

Another helpful data point would be subscriber count at the time of upload, since creators with larger existing audiences tend to see higher initial view counts. Thumbnail performance and click-through rate would also help explain why some videos get more views, as a more appealing thumbnail can lead to more people clicking on the video. Finally, information on upload timing, such as the day and time a video is published, could help explain differences in performance.

Use this chart table for question 13!

This histogram shows how many videos have a certain range of views, like how many videos have between 0 and 1,000 views, how many have between 1,000 and 5,000 views, and so on.

